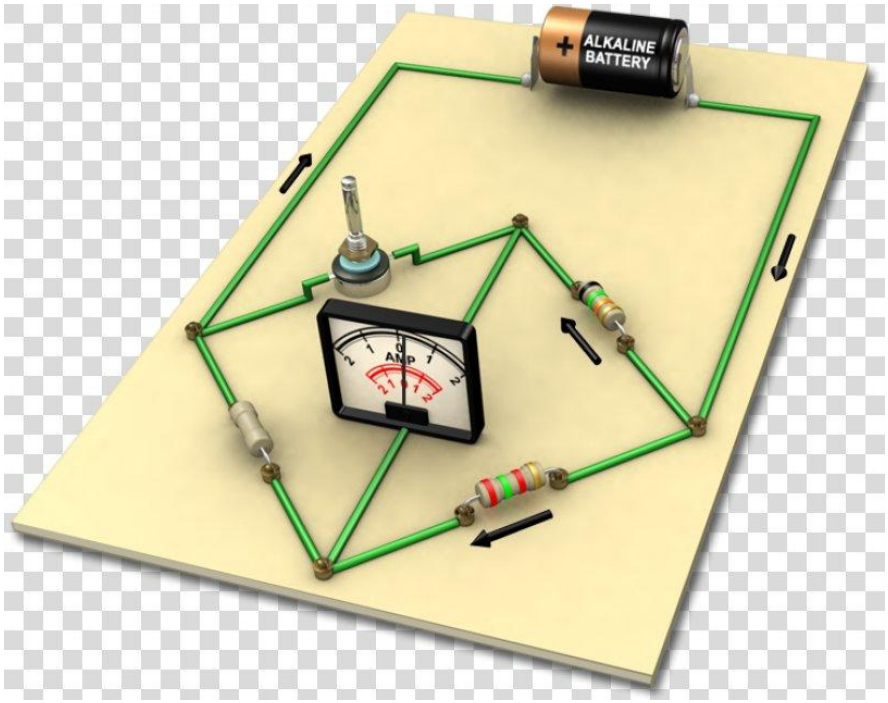
EPM213s - Electrical Measurements

Exp 3 Bridges

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Spring 25

Indirect Measurements of Resistance Using Bridges

1. objective

The objective of this experiment is to determine the resistance of an unknown resistor using bridge circuits, such as the Wheatstone Bridge and Kelvin Bridge, which employ indirect measurement techniques. These bridges yield precise resistance values by balancing voltages and currents instead of directly measuring resistance.

2. procedure

Select appropriate values for P and Q from the laboratory panel.

Use the variable (0 → 1000 ohm) resistor as the standard arm, with R as the unknown resistance.

Connect the bridge as illustrated in Figure (1).

Adjust the value of S until balance is achieved, then determine R using the Wheatstone bridge equations.

Modify the values of P and Q and repeat the procedure.

Measure R using an AVO meter and compare it with the calculated Wheatstone bridge results.

Connect the AC Bridge as discussed in the lab.

Adjust the variable resistance until balance is reached.

Measure the values of all elements in the bridge arms. Confirm the balance condition.

5. Photos

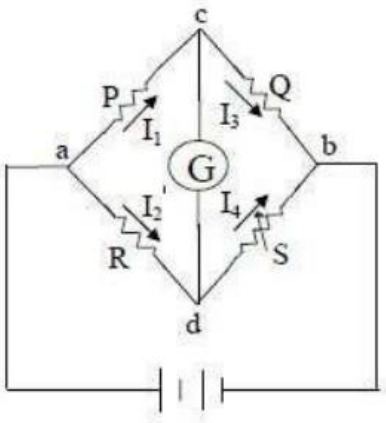
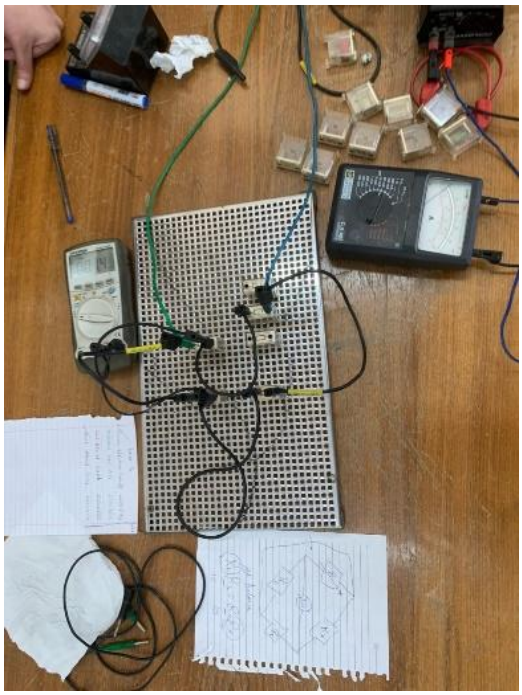
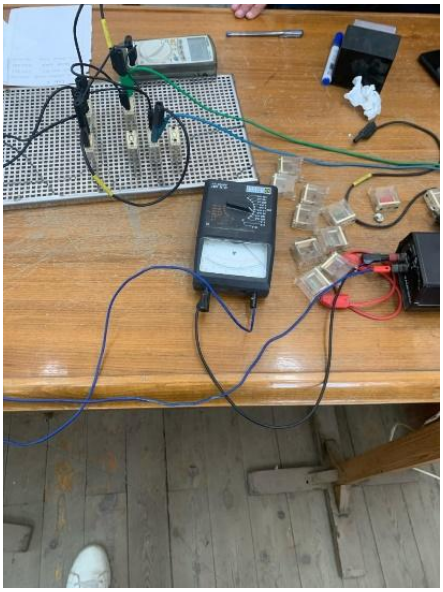


Figure 1. Whetstone Bridge.





Conclusion

The indirect measurement of resistance using bridge circuits ensures high accuracy by balancing voltages rather than depending on direct ohmmeter readings. The Wheatstone Bridge is effective for medium-range resistances, whereas the Kelvin Bridge is well-suited for measuring very low resistances.